FATHOM®

Version	Revision Date:	SDS Number:	Date of last issue: -
1.1	25.08.2025	50001830	Date of first issue: 01.06.2018

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name FATHOM®

Other means of identification

Product code 50001830

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Substance/Mixture : Fungicide

Recommended restrictions on use : Use as recommended by the label.

1.3 Details of the supplier of the safety data sheet

Supplier Address

FMC Agro Limited
Rectors Lane, Pentre
Flintshire
CH5 2DH
United Kingdom

Telephone: + 44 1244 537370
E-mail address: SDS-Info@fmc.com .

1.4 Emergency telephone number

For leak, fire, spill or accident emergencies, call:
England and Wales: 44-870-8200418 (CHEMTREC)

Medical emergency:
England and Wales: 111
Scotland: 84 54 24 2424

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008) as amended by GB-CLP Regulation, UK SI 2019/720, and UK SI 2020/1567)

Skin sensitisation, Sub-category 1B H317: May cause an allergic skin reaction.

Eye irritation, Category 2	H319: Causes serious eye irritation.
Reproductive toxicity, Category 2	H361d: Suspected of damaging the unborn child.
Long-term (chronic) aquatic hazard, Category 1	H410: Very toxic to aquatic life with long lasting effects.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008) as amended by GB-CLP Regulation, UK SI 2019/720, and UK SI 2020/1567)

Hazard pictograms :

Signal word : Warning

Hazard statements :

H317 May cause an allergic skin reaction.
H319 Causes serious eye irritation.
H361d Suspected of damaging the unborn child.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary statements :

Prevention:

P201 Obtain special instructions before use.
P261 Avoid breathing mist or vapours.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response:

P308 + P313 IF exposed or concerned: Get medical advice/ attention.
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Disposal:

P501 Dispose of contents/container as hazardous waste in accordance with local regulations.

Hazardous components which must be listed on the label:
tebuconazole (ISO)

Additional Labelling

EUH401 To avoid risks to human health and the environment, comply with the instructions for use.

For special phrases (SP) and safety intervals, consult the label.

2.3 Other hazards

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number	Classification	Concentration (% w/w)
tebuconazole (ISO)	107534-96-3 403-640-2 603-197-00-7	Acute Tox. 4; H302 Repr. 2; H361d Aquatic Acute 1; H400 Aquatic Chronic 1; H410 M-Factor (Acute aquatic toxicity): 1 M-Factor (Chronic aquatic toxicity): 10	>= 10 - < 20
octan-1-ol	111-87-5 203-917-6	Acute Tox. 4; H302 Acute Tox. 4; H312 Eye Irrit. 2; H319 Aquatic Chronic 3; H412	>= 10 - < 20
Tristyryl phenol-polyethylene glycol-phosphoric acid ester	114535-82-9	Eye Irrit. 2; H319 Aquatic Chronic 3; H412	>= 2.5 - < 10
Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs.	85536-14-7 287-494-3	Acute Tox. 4; H302 Skin Corr. 1C; H314 Eye Dam. 1; H318 Aquatic Chronic 3; H412	>= 1 - < 2.5

For explanation of abbreviations see section 16.

SECTION 4: First aid measures

4.1 Description of first aid measures

General advice	:	Move out of dangerous area. Show this safety data sheet to the doctor in attendance. Do not leave the victim unattended.
Protection of first-aiders	:	Avoid inhalation, ingestion and contact with skin and eyes.
If inhaled	:	Remove to fresh air. If unconscious, place in recovery position and seek medical advice. If experiencing any discomfort, immediately remove from exposure. Light cases: Keep person under surveillance. Get medical attention immediately if symptoms develop. Serious cases: Get medical attention immediately or call for an ambulance.
In case of skin contact	:	If on clothes, remove clothes. If on skin, rinse well with water. Wash off with soap and plenty of water. Get medical attention immediately if irritation develops and persists.
In case of eye contact	:	Immediately flush eye(s) with plenty of water. Remove contact lenses. Protect unharmed eye. Keep eye wide open while rinsing. If eye irritation persists, consult a specialist.
If swallowed	:	Keep respiratory tract clear. Do not give milk or alcoholic beverages. Never give anything by mouth to an unconscious person. Take victim immediately to hospital. Do not induce vomiting without medical advice.

4.2 Most important symptoms and effects, both acute and delayed

Symptoms	:	The first symptom to appear after skin or eye contact will be irritation. After ingestion, the main symptoms are passivity, impaired mobility and shortness of breath.
Risks	:	May cause an allergic skin reaction. Causes serious eye irritation. Suspected of damaging the unborn child.

4.3 Indication of any immediate medical attention and special treatment needed

Treatment	:	Treat symptomatically.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media	:	Dry chemical, CO2, water spray or regular foam. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	:	Do not spread spilled material with high-pressure water streams. High volume water jet

5.2 Special hazards arising from the substance or mixture

Specific hazards during firefighting	:	Do not allow run-off from fire fighting to enter drains or water courses.
Hazardous combustion products	:	Fire may produce irritating, corrosive and/or toxic gases. Carbon oxides Nitrogen oxides (NOx) Sulphur oxides Hydrogen chloride Oxides of phosphorus Chlorinated compounds

5.3 Advice for firefighters

Special protective equipment for firefighters	:	Wear self-contained breathing apparatus for firefighting if necessary.
Further information	:	Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. For safety reasons in case of fire, cans should be stored separately in closed containments. Use a water spray to cool fully closed containers.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions	:	Evacuate personnel to safe areas. Use personal protective equipment. If it can be safely done, stop the leak. Do not touch or walk through the spilled material. Never return spills in original containers for re-use. Mark the contaminated area with signs and prevent access to unauthorized personnel. Only qualified personnel equipped with suitable protective equipment may intervene.

6.2 Environmental precautions

Environmental precautions : Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform respective authorities.

6.3 Methods and material for containment and cleaning up

Methods for cleaning up : Neutralize with chalk, alkali solution or ammonia.
Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).
Keep in suitable, closed containers for disposal.

6.4 Reference to other sections

See sections: 7, 8, 11, 12 and 13.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advice on safe handling : Avoid formation of aerosol.
Do not breathe vapours/dust.
Avoid exposure - obtain special instructions before use.
Avoid contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Provide sufficient air exchange and/or exhaust in work rooms.
Dispose of rinse water in accordance with local and national regulations.
Persons susceptible to skin sensitisation problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.

Advice on protection against fire and explosion : Do not spray on a naked flame or any incandescent material.
Keep away from open flames, hot surfaces and sources of ignition.

Hygiene measures : When using do not eat or drink. When using do not smoke.
Wash hands before breaks and at the end of workday. Remove and wash contaminated clothing and gloves, including the inside, before re-use.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage : No smoking. Keep in a well-ventilated place. Containers which

areas and containers		are opened must be carefully resealed and kept upright to prevent leakage. Observe label precautions. Electrical installations / working materials must comply with the technological safety standards.
Further information on storage conditions	:	The product is stable under normal conditions of warehouse storage. Protect from frost and extreme heat. At temperatures below -10°C crystallisation may occur. The product is degraded by fluorinated packaging materials. Store in closed, labelled containers. The storage room should be constructed of incombustible material, closed, dry, ventilated and with impermeable floor, without access of unauthorised persons or children. A warning sign reading “POISON” is recommended. The room should only be used for storage of chemicals. Food, drink, feed and seed should not be present. A hand wash station should be available.
Recommended storage temperature	:	5 - 30 °C
Further information on storage stability	:	No decomposition if stored and applied as directed.

7.3 Specific end use(s)

Specific use(s)	:	Registered pesticide to be used in accordance with a label approved by country-specific regulatory authorities.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational Exposure Limits

Contains no substances with occupational exposure limit values.

Derived No Effect Level (DNEL)

Substance name	End Use	Exposure routes	Potential health effects	Value
Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs.	Workers	Inhalation	Long-term systemic effects	6 mg/m3
	Workers	Dermal	Long-term systemic effects	85 mg/kg bw/day
	Consumers	Inhalation	Long-term systemic effects	1.5 mg/m3
	Consumers	Dermal	Long-term systemic effects	42.5 mg/kg bw/day
	Consumers	Oral	Long-term systemic	0.425 mg/kg

			effects	bw/day

Predicted No Effect Concentration (PNEC)

Substance name	Environmental Compartment	Value
octan-1-ol	Fresh water	200 µg/l
	Marine water	20 µg/l
	Sewage treatment plant	55.5 mg/l
	Fresh water sediment	2.1 mg/kg dry weight (d.w.)
	Marine sediment	0.210 mg/kg dry weight (d.w.)
	Soil	1.6 mg/kg dry weight (d.w.)
Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs.	Fresh water	0.268 mg/l
	Marine water	0.027 mg/l
	Fresh water sediment	8.1 mg/kg dry weight (d.w.)
	Marine sediment	6.8 mg/kg dry weight (d.w.)
	Soil	35 mg/kg dry weight (d.w.)
	Intermittent use (freshwater)	0.0167 mg/l
	Sewage treatment plant	3.43 mg/l

8.2 Exposure controls

Personal protective equipment

Eye/face protection : Eye wash bottle with pure water
Tightly fitting safety goggles
Wear face-shield and protective suit for abnormal processing problems.

Hand protection
Material : Wear chemical resistant gloves, such as barrier laminate, butyl rubber or nitrile rubber.

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Respiratory protection : In case of mist, spray or aerosol exposure wear suitable personal respiratory protection and protective suit.

Protective measures : Plan first aid action before beginning work with this product.
Always have on hand a first-aid kit, together with proper instructions.
Wear suitable protective equipment.
When using do not eat, drink or smoke.

In the context of professional plant protection use as recommended, the end user must refer to the label and the instructions for use.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	:	liquid
Colour	:	light yellow
Odour	:	like soap
Odour Threshold	:	not determined
pH	:	3.5 (25 °C) Concentration: 1 %
Melting point/freezing point	:	not determined
Boiling point/boiling range	:	not determined
Flash point	:	73 °C Method: closed cup
Evaporation rate	:	not determined
Vapour pressure	:	Not available for this mixture.
Relative vapour density	:	not determined
Density	:	978 g/l (20 °C)
Solubility(ies)		
Water solubility	:	No data available
Solubility in other solvents	:	No data available
Partition coefficient: n-octanol/water	:	Not available for this mixture.
Decomposition temperature	:	not determined
Viscosity		
Viscosity, dynamic	:	8.99 mPa,s (20 °C) 4.90 mPa,s (40 °C)
Explosive properties	:	Not explosive
Oxidizing properties	:	Non-oxidizing

9.2 Other information

Flammability (liquids)	:	No applicable data available.
Particle size	:	Not applicable
Particle Size Distribution	:	Not applicable
Self-ignition	:	262 °C

SECTION 10: Stability and reactivity

10.1 Reactivity

No decomposition if stored and applied as directed.

10.2 Chemical stability

No decomposition if stored and applied as directed.

10.3 Possibility of hazardous reactions

Hazardous reactions : No decomposition if stored and applied as directed.

Vapours may form explosive mixture with air.

10.4 Conditions to avoid

Conditions to avoid : Heat, flames and sparks.
Protect from frost, heat and sunlight.
Heating of the product will produce harmful and irritant vapours.

10.5 Incompatible materials

Materials to avoid : Avoid strong acids, bases, and oxidizers

10.6 Hazardous decomposition products

Stable under recommended storage conditions.

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Acute toxicity

Product:

Acute oral toxicity : LD50 (Rat): > 2,000 mg/kg
Method: OECD Test Guideline 420

Acute inhalation toxicity : LC50 (Rat): > 5.13 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403

Acute dermal toxicity : LD50 (Rat): > 2,000 mg/kg

Components:

tebuconazole (ISO):

Acute oral toxicity : LD50 (Rat, female): > 2,000 mg/kg
Method: OECD Test Guideline 425
Symptoms: ataxia, Lethargy, Breathing difficulties

		Assessment: The component/mixture is minimally toxic after single ingestion. Remarks: no mortality
Acute inhalation toxicity	:	LC50 (Rat, male and female): > 5.18 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 GLP: yes Remarks: no mortality
Acute dermal toxicity	:	LD50 (Rat): > 2,000 mg/kg Method: OECD Test Guideline 402 GLP: yes Assessment: The component/mixture is minimally toxic after single contact with skin. Remarks: no mortality

octan-1-ol:

Acute oral toxicity	:	LD50 (Rat, male): 1,800 mg/kg LD50 (Rat, female): 720 mg/kg
Acute inhalation toxicity	:	LC50 (Rat): > 2.05 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: US EPA Test Guideline OPPTS 870.1300 Assessment: The substance or mixture has no acute inhalation toxicity
Acute dermal toxicity	:	LD50 (Rabbit, male and female): > 1,500 - < 2,000 mg/kg

Tristyryl phenol-polyethylene glycol-phosphoric acid ester:

Acute oral toxicity	:	LD50 (Rat): > 2,000 mg/kg Method: OECD Test Guideline 401

Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs.:

Acute oral toxicity	:	LD50 Oral (Rat, male and female): 1,470 mg/kg Method: OECD Test Guideline 401 Symptoms: Diarrhoea, ataxia, diuresis, Tremors, dryness of the eyes Remarks: mortality
Acute dermal toxicity	:	LD50 Dermal (Rat, male and female): > 2,000 mg/kg Method: OECD Test Guideline 402 Remarks: no mortality

Skin corrosion/irritation

Product:

Assessment	:	No skin irritation
Method	:	OECD Test Guideline 404
Remarks	:	May cause mild irritation. Minimal effects that do not meet the threshold for classification.

Components:

tebuconazole (ISO):

Species	:	Rabbit
Assessment	:	Not classified as irritant
Method	:	OECD Test Guideline 404
Result	:	slight irritation
GLP	:	yes

octan-1-ol:

Species	:	Rabbit
Method	:	OECD Test Guideline 404
Result	:	Mild skin irritation

Tristyryl phenol-polyethylene glycol-phosphoric acid ester:

Species	:	Rabbit
Method	:	OECD Test Guideline 404
Result	:	No skin irritation

Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs.:

Species	:	Rabbit
Method	:	OECD Test Guideline 404
Result	:	Corrosive after 1 to 4 hours of exposure

Serious eye damage/eye irritation

Product:

Assessment	:	Irritating to eyes.
Method	:	OECD Test Guideline 405
Result	:	Irritation to eyes, reversing within 21 days

Components:

tebuconazole (ISO):

Species	:	Rabbit
Assessment	:	No eye irritation
Method	:	FIFRA 81.04
Remarks	:	Minimal effects that do not meet the threshold for classification.

octan-1-ol:

Species : Rabbit
Method : OECD Test Guideline 405
Result : Irritation to eyes, reversing within 21 days

Tristyryl phenol-polyethylene glycol-phosphoric acid ester:

Species : Rabbit
Method : OECD Test Guideline 405
Result : Eye irritation

Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs.:

Species : Rabbit
Method : OECD Test Guideline 405
Result : Irreversible effects on the eye

Respiratory or skin sensitisation

Product:

Species : Mouse
Method : OECD Test Guideline 429
Result : The product is a skin sensitiser, sub-category 1B.

Components:

tebuconazole (ISO):

Method : OECD Test Guideline 406
Result : Not a skin sensitizer.

Test Type : Local lymph node assay (LLNA)
Exposure routes : Skin contact
Species : Mouse
Method : OECD Test Guideline 429
Result : Not a skin sensitizer.

octan-1-ol:

Test Type : Maximisation Test
Species : Guinea pig
Method : OECD Test Guideline 406
Result : Does not cause skin sensitisation.
Remarks : Based on data from similar materials

Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs.:

Test Type : Maximisation Test
Species : Guinea pig
Result : Does not cause skin sensitisation.

Germ cell mutagenicity

Components:

tebuconazole (ISO):

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Result: negative

octan-1-ol:

Genotoxicity in vitro : Test Type: In vitro mammalian cell gene mutation test
Method: OECD Test Guideline 476
Result: negative

Test Type: reverse mutation assay
Method: OECD Test Guideline 471
Result: negative

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse (male and female)
Application Route: Oral
Method: OECD Test Guideline 474
Result: negative

Germ cell mutagenicity- Assessment : Weight of evidence does not support classification as a germ cell mutagen.

Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs.:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: Regulation (EC) No. 440/2008, Annex, B.13/14 (Ames test)
Result: negative

Test Type: Chromosome aberration test in vitro
Method: OECD Test Guideline 473
Result: equivocal

Test Type: In vitro mammalian cell gene mutation test
Method: OECD Test Guideline 476
Result: negative
Remarks: Based on data from similar materials

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse (male and female)
Method: OECD Test Guideline 474
Result: negative

Test Type: Cytogenetic assay
Species: Rat (male)
Result: negative

		Test Type: Rodent Dominant Lethal Assay Species: Mouse (male) Result: negative
Germ cell mutagenicity- Assessment	:	Weight of evidence does not support classification as a germ cell mutagen.

Carcinogenicity

Reproductive toxicity

Product:

Reproductive toxicity - Assessment	:	Suspected of damaging the unborn child.

Components:

octan-1-ol:

Effects on fertility	:	Test Type: one-generation reproductive toxicity Species: Rat, male and female Application Route: Oral Dose: 10, 100, 1000 mg/kg bw/day General Toxicity - Parent: NOAEL: 1,000 mg/kg bw/day General Toxicity F1: NOAEL: 1,000 mg/kg bw/day Result: negative
Effects on foetal development	:	Species: Rat Application Route: Oral Dose: 0,130,650,975,1300 mg/kg bw/day Duration of Single Treatment: 20 d General Toxicity Maternal: LOAEL: 650 mg/kg bw/day Embryo-foetal toxicity: NOAEL: 1,300 mg/kg bw/day Symptoms: Maternal effects Method: OECD Test Guideline 414
Reproductive toxicity - Assessment	:	Weight of evidence does not support classification for reproductive toxicity

Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs.:

Effects on fertility	:	Test Type: Three-generation study Species: Rat, male and female Application Route: Oral Dose: 14, 70, 350mg/kg bw d Duration of Single Treatment: 2 yr General Toxicity - Parent: NOAEL: 350 mg/kg bw/day General Toxicity F1: NOAEL: 350 mg/kg bw/day General Toxicity F2: NOAEL: 350 mg/kg bw/day Result: negative

Effects on foetal development	:	Test Type: reproductive and developmental toxicity study Species: Rat Application Route: Oral Dose: 0.2, 2, 300, 600 milligram per kilogram Duration of Single Treatment: 20 d General Toxicity Maternal: NOAEL: 300 mg/kg bw/day Embryo-foetal toxicity: NOAEL: 300 mg/L Symptoms: Diarrhoea, Reduced body weight, Retardations Result: negative Remarks: Based on data from similar materials
		Test Type: reproductive and developmental toxicity study Species: Mouse Application Route: Oral Dose: 0.2, 2, 300, 600mg/kg/bw General Toxicity Maternal: NOAEL: 2 mg/kg bw/day Embryo-foetal toxicity: NOAEL: 300 mg/kg bw/day Remarks: Based on data from similar materials
Reproductive toxicity - Assessment	:	Weight of evidence does not support classification for reproductive toxicity

STOT - single exposure

STOT - repeated exposure

<u>Components:</u>

octan-1-ol:

Assessment	:	The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs.:

Assessment	:	The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity

<u>Components:</u>

octan-1-ol:

Species	:	Rat, male
NOAEL	:	1127 mg/kg bw/day
Application Route	:	Oral
Exposure time	:	13 Weeks
Dose	:	182, 374, 1127 mg/kg bw/day

Species	:	Rat, female
NOAEL	:	1243 mg/kg bw/day
Application Route	:	Oral
Exposure time	:	13 Weeks

Dose	:	216, 427, 1243 mg/kg bw/day

Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs.:

Species	:	Rat, male and female
NOAEL	:	85 mg/kg bw/day
LOAEL	:	300 mg/kg bw/day
Application Route	:	Oral
Exposure time	:	9 months
Dose	:	300, 900mg/kg/bw/day
Target Organs	:	Kidney, Liver
Symptoms	:	kidney effects, Liver effects
Remarks	:	Based on data from similar materials

Species	:	Rat, male and female
NOAEL	:	5 %
Application Route	:	Dermal
Exposure time	:	26 weeks
Dose	:	0.5, 1, 5 %
Remarks	:	Based on data from similar materials

Aspiration toxicity

Further information

Product:

Remarks	:	The first symptom to appear after skin or eye contact will be irritation. After ingestion, the main symptoms are passivity, impaired mobility and shortness of breath.

Components:

tebuconazole (ISO):

Remarks	:	The main symptoms were passivity, impaired mobility and shortness of breath at high doses in animal tests.

SECTION 12: Ecological information

12.1 Toxicity

Product:

Toxicity to fish	:	LC50 (Oncorhynchus mykiss (rainbow trout)): 24.2 mg/l Exposure time: 96 h
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 17.2 mg/l Exposure time: 48 h
Toxicity to algae/aquatic plants	:	EC50 (Desmodesmus subspicatus (green algae)): 28.05 mg/l Exposure time: 72 h

		NOEC (Desmodesmus subspicatus (green algae)): 2.88 mg/l Exposure time: 72 h
Toxicity to soil dwelling organisms	:	LC50: 1,203 mg/kg Exposure time: 14 d Species: Eisenia fetida (earthworms)
Toxicity to terrestrial organisms	:	LD50: 74 µg/bee Exposure time: 48 h End point: Acute oral toxicity Species: Apis mellifera (bees) LD50: 339 µg/bee Exposure time: 48 h End point: Acute contact toxicity Species: Apis mellifera (bees)

Ecotoxicology Assessment

Chronic aquatic toxicity	:	Very toxic to aquatic life with long lasting effects.

Components:

tebuconazole (ISO):

Toxicity to fish	:	LC50 (Oncorhynchus mykiss (rainbow trout)): 4.4 mg/l Exposure time: 96 h Test Type: flow-through test LC50 (Lepomis macrochirus (Bluegill sunfish)): 5.7 mg/l Exposure time: 96 h LC50 (Leuciscus idus (Golden orfe)): 8.7 mg/l Exposure time: 96 h
Toxicity to daphnia and other aquatic invertebrates	:	LC50 (Daphnia magna (Water flea)): 2.79 mg/l Exposure time: 48 h Test Type: flow-through test
Toxicity to algae/aquatic plants	:	ErC50 (Pseudokirchneriella subcapitata (algae)): 3.8 mg/l Exposure time: 72 h Test Type: static test ErC50 (Scenedesmus quadricauda (Green algae)): 5.3 mg/l Exposure time: 72 h EC50 (Lemna gibba (duckweed)): 0.144 mg/l Exposure time: 14 d
M-Factor (Acute aquatic toxicity)	:	1
Toxicity to fish (Chronic tox-	:	NOEC: 0.012 mg/l

icity)		Exposure time: 60 d Species: Salmo gairdneri
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC: 0.12 mg/l Exposure time: 21 d Species: Daphnia magna (Water flea)
M-Factor (Chronic aquatic toxicity)	:	10
Toxicity to soil dwelling organisms	:	LC50: 1,381 mg/kg Exposure time: 14 d Species: Eisenia fetida (earthworms)
Toxicity to terrestrial organisms	:	LD50: 1,988 mg/kg Species: Colinus virginianus (Bobwhite quail) LD50: > 200 µg/bee Species: Apis mellifera (bees) Remarks: Contact LD50: > 83 µg/bee Exposure time: 48 h Species: Apis mellifera (bees) LD50: 2,912 mg/kg Species: Coturnix japonica (Japanese quail)

octan-1-ol:

Toxicity to fish	:	LC50 (Pimephales promelas (fathead minnow)): 13.3 mg/l Exposure time: 96 h Test Type: flow-through test
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 20 mg/l Exposure time: 24 h Method: OECD Test Guideline 202
Toxicity to algae/aquatic plants	:	EC10 (Desmodesmus subspicatus (green algae)): 4.2 mg/l Exposure time: 48 h Test Type: static test EC50 (Desmodesmus subspicatus (green algae)): 6.5 mg/l Exposure time: 48 h Test Type: static test
Toxicity to microorganisms	:	(Protozoa): 44 mg/l Exposure time: 72 h Test Type: Cell multiplication inhibition test Remarks: Based on data from similar materials
Toxicity to daphnia and other	:	NOEC: 1 mg/l

aquatic invertebrates (Chronic toxicity)		Exposure time: 21 d Species: Daphnia magna (Water flea) Method: OECD Test Guideline 211

Tristyryl phenol-polyethylene glycol-phosphoric acid ester:

Toxicity to fish	:	LC50 (Leuciscus idus (Golden orfe)): 100 - 500 mg/l Exposure time: 96 h
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): > 100 mg/l Exposure time: 48 h Method: OECD Test Guideline 202
Toxicity to algae/aquatic plants	:	NOEC (Desmodesmus subspicatus (green algae)): > 100 mg/l Exposure time: 72 h Method: OECD Test Guideline 201 EC50 (Desmodesmus subspicatus (green algae)): > 100 mg/l Exposure time: 72 h Method: OECD Test Guideline 201

Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs.:

Toxicity to fish	:	LC50 (Lepomis macrochirus (Bluegill sunfish)): 1.67 mg/l Exposure time: 96 h Method: OPPTS 850.1075
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 2.9 mg/l Exposure time: 48 h Method: OECD Test Guideline 202 Remarks: Based on data from similar materials
Toxicity to algae/aquatic plants	:	EC50 (Pseudokirchneriella subcapitata (green algae)): 235 mg/l Exposure time: 72 h Method: OECD Test Guideline 201 Remarks: Based on data from similar materials NOEC : >= 4 mg/l Exposure time: 28 d Test Type: flow-through test Remarks: Based on data from similar materials
Toxicity to fish (Chronic toxicity)	:	NOEC: 0.23 mg/l Exposure time: 72 d Species: Oncorhynchus mykiss (rainbow trout) Test Type: flow-through test Method: OECD Test Guideline 210 Remarks: Based on data from similar materials
Toxicity to daphnia and other aquatic invertebrates (Chron-	:	NOEC: 1.18 mg/l Exposure time: 21 d

ic toxicity)		Species: Daphnia magna (Water flea) Test Type: flow-through test Method: OECD Test Guideline 211 Remarks: Based on data from similar materials
Toxicity to soil dwelling organisms	:	NOEC: 250 mg/kg Exposure time: 14 d Species: Eisenia fetida (earthworms) Method: OECD Test Guideline 207 Remarks: Based on data from similar materials

12.2 Persistence and degradability

Product:

Biodegradability : Remarks: Product contains minor amounts of not readily biodegradable components, which may not be degradable in waste water treatment plants.

Components:

tebuconazole (ISO):

Biodegradability : Result: Not readily biodegradable.

octan-1-ol:

Biodegradability : Inoculum: activated sludge
Result: Readily biodegradable.
Biodegradation: 82.2 %
Exposure time: 28 d
Method: OECD Test Guideline 301B

Tristyryl phenol-polyethylene glycol-phosphoric acid ester:

Biodegradability : Result: Not readily biodegradable.
Biodegradation: 30 - 40 %
Method: OECD Test Guideline 302B

Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs.:

Biodegradability : Inoculum: activated sludge
Result: Readily biodegradable.
Method: OECD Test Guideline 301A

12.3 Bioaccumulative potential

Product:

Bioaccumulation : Remarks: No data is available on the product itself.

Components:

tebuconazole (ISO):

Bioaccumulation	:	Species: Fish Bioconcentration factor (BCF): 65 Remarks: Bioaccumulation is unlikely.
Partition coefficient: n-octanol/water	:	log Pow: 3.7 (20 °C)

octan-1-ol:

Partition coefficient: n-octanol/water	:	log Pow: 3.5 (23 °C) pH: 5.7

Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs.:

Bioaccumulation	:	Species: Pimephales promelas (fathead minnow) Bioconcentration factor (BCF): 6.0 Method: OECD Test Guideline 305A Remarks: Based on data from similar materials
Partition coefficient: n-octanol/water	:	log Pow: 2.2 (23 °C) pH: 3.7

12.4 Mobility in soil

Product:

Distribution among environmental compartments	:	Remarks: No data is available on the product itself.

Components:

tebuconazole (ISO):

Distribution among environmental compartments	:	Remarks: Low mobility in soil

12.5 Results of PBT and vPvB assessment

Product:

Assessment	:	This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

12.6 Other adverse effects

Product:

Endocrine disrupting potential	:	This substance/mixture does not contain components considered to have endocrine disrupting properties for environment according to UK REACH Article 57(f).
Additional ecological infor-	:	An environmental hazard cannot be excluded in the event of

mation		unprofessional handling or disposal. Very toxic to aquatic life with long lasting effects.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product	:	The product should not be allowed to enter drains, water courses or the soil. Do not contaminate ponds, waterways or ditches with chemical or used container. Send to a licensed waste management company.
Contaminated packaging	:	Empty remaining contents. Triple rinse containers. Do not re-use empty containers. Packaging that is not properly emptied must be disposed of as the unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal.

SECTION 14: Transport information

14.1 UN number

ADN	:	UN 3082
ADR	:	UN 3082
RID	:	UN 3082
IMDG	:	UN 3082
IATA	:	UN 3082

14.2 UN proper shipping name

ADN	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Tebuconazole)
ADR	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Tebuconazole)
RID	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Tebuconazole)
IMDG	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Tebuconazole)

IATA	:	Environmentally hazardous substance, liquid, n.o.s. (Tebuconazole)

14.3 Transport hazard class(es)

		Class	Subsidiary risks
ADN	:	9	
ADR	:	9	
RID	:	9	
IMDG	:	9	
IATA	:	9	

14.4 Packing group

ADN

Packing group	:	III
Classification Code	:	M6
Hazard Identification Number	:	90
Labels	:	9

ADR

Packing group	:	III
Classification Code	:	M6
Hazard Identification Number	:	90
Labels	:	9
Tunnel restriction code	:	(-)

RID

Packing group	:	III
Classification Code	:	M6
Hazard Identification Number	:	90
Labels	:	9

IMDG

Packing group	:	III
Labels	:	9
EmS Code	:	F-A, S-F

IATA (Cargo)

Packing instruction (cargo aircraft)	:	964
Packing instruction (LQ)	:	Y964
Packing group	:	III
Labels	:	Miscellaneous

IATA (Passenger)

Packing instruction (passenger aircraft)	:	964
Packing instruction (LQ)	:	Y964
Packing group	:	III
Labels	:	Miscellaneous

14.5 Environmental hazards

ADN
Environmentally hazardous : yes

ADR
Environmentally hazardous : yes

RID
Environmentally hazardous : yes

IMDG
Marine pollutant : yes

IATA (Passenger)
Environmentally hazardous : yes

IATA (Cargo)
Environmentally hazardous : yes

14.6 Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

14.7 Transport in bulk according to Annex II of Marpol and the IBC Code

Not applicable for product as supplied.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Relevant EU provisions transposed through retained EU law

UK REACH List of restrictions (Annex 17)	:	Conditions of restriction for the following entries should be considered: Number on list 3 octan-1-ol (Number on list 3) Tristyryl phenol-polyethylene glycol-phosphoric acid ester (Number on list 3) Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs. (Number on list 3)
UK REACH Candidate list of substances of very high concern (SVHC) for Authorisation	:	Not applicable
The Persistent Organic Pollutants Regulations (retained Regulation (EU) 2019/1021 as amended for Great Britain)	:	Not applicable
Regulation (EU) No 2024/590 on substances that deplete the ozone layer	:	Not applicable

UK REACH List of substances subject to authorisation (Annex XIV)	:	Not applicable
Control of Major Accident Hazards Regulations 2015 (COMAH)	E1	ENVIRONMENTAL HAZARDS

Other regulations:

Take note of The Management of Health and Safety at Work Regulations 1999 (requirements relating to new and expectant mothers at work contained in Regulation 16 to 18) and of the Pregnant Workers Directive 92/85/EEC.

Take note of The Management of Health and Safety at Work Regulations 1999 (requirements relating to protection of young people at work contained in Regulation 19) and of Directive 94/33/EC on the protection of young people at work.

The components of this product are reported in the following inventories:

TCSI	:	On the inventory, or in compliance with the inventory
TSCA	:	Product contains substance(s) not listed on TSCA inventory.
AIIC	:	Not in compliance with the inventory
DSL	:	This product contains the following components that are not on the Canadian DSL nor NDSL. tebuconazole (ISO)
ENCS	:	Not in compliance with the inventory
ISHL	:	Not in compliance with the inventory
KECI	:	Not in compliance with the inventory
PICCS	:	Not in compliance with the inventory
IECSC	:	Not in compliance with the inventory
NZIoC	:	Not in compliance with the inventory
TECI	:	Not in compliance with the inventory

15.2 Chemical safety assessment

A chemical safety assessment is not required for this product (mixture).

SECTION 16: Other information

Full text of H-Statements

H302	:	Harmful if swallowed.

H312	:	Harmful in contact with skin.
H314	:	Causes severe skin burns and eye damage.
H318	:	Causes serious eye damage.
H319	:	Causes serious eye irritation.
H361d	:	Suspected of damaging the unborn child.
H400	:	Very toxic to aquatic life.
H410	:	Very toxic to aquatic life with long lasting effects.
H412	:	Harmful to aquatic life with long lasting effects.

Full text of other abbreviations

Acute Tox.	:	Acute toxicity
Aquatic Acute	:	Short-term (acute) aquatic hazard
Aquatic Chronic	:	Long-term (chronic) aquatic hazard
Eye Dam.	:	Serious eye damage
Eye Irrit.	:	Eye irritation
Repr.	:	Reproductive toxicity
Skin Corr.	:	Skin corrosion

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of very high concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative

Further information

Other information :

Classification of the mixture:		**Classification procedure:**
Skin Sens. 1B	H317	Based on product data or assessment
Eye Irrit. 2	H319	Based on product data or assessment
Repr. 2	H361d	Calculation method
Aquatic Chronic 1	H410	Based on product data or assessment

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